Lab Assignment 3 > lab_assignment3.py

```python
import time

#Ivan E Garcia eaa108

def fibonacci_naive(n):
    if n <= 1:
        return n

    return fibonacci_naive(n - 2) + fibonacci_naive(n - 1)

def fibonacci_memo(n, memo={}):
    if n <= 1:
        return n
    if n in memo:
        return memo[n]
    memo[n] = fibonacci_memo(n - 2, memo) + fibonacci_memo(n - 1, memo)

    return memo[n]


start_time = time.time()
fibonacci_naive(40)
end_time = time.time()
elapsed_time = end_time - start_time

print("Naive time:", elapsed_time, "sec")


start_time = time.time()
fibonacci_memo(40)
end_time = time.time()
elapsed_time = end_time - start_time

print("Memo time :", elapsed_time, "sec")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[ivan@R-3D Lab Assignment 3]$ python3 lab_assignment3.oy
python3: can't open file '/home/ivan/Documents/personal-stuff/UTSA Stuff/UTSA-Coursework/Fall 2026/Systems programming/Lab Assignment 3/lab_assignment3.oy': [Errno 2] No such file or directory
[ivan@R-3D Lab Assignment 3]$ python3 lab_assignment3.py
Naive time: 6.931756258010864 sec
Memo time : 4.0531158447265625e-05 sec
[ivan@R-3D Lab Assignment 3]$
```